

32.6 Bayesian thread III: mixed models as hierarchical models

The Bayesian thread began in [Section 7.6](#) with a conjugate prior on $(\boldsymbol{\beta}, \sigma^2)$ and continued in [Section 12.6](#) with credible sets and predictive distributions. Mixed models are where the two traditions meet most closely, because a random effect *is* a prior: the statement $\mathbf{u} \sim N_q(\mathbf{0}, \mathbf{G})$ is formally indistinguishable from a normal prior on q unknown constants. What separates the methods of [Section 32.4](#) from a fully Bayesian analysis is only that they estimate $\mathbf{G}$ and then treat it as known, which is empirical Bayes (Proposition 27.5.4), while a Bayesian puts a prior on $\mathbf{G}$ too and integrates.

32.6.1 The hierarchical reading

Write the model in three levels:

$$\mathbf{Y} \mid \boldsymbol{\beta}, \mathbf{u}, \boldsymbol{\theta} \sim N_n(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}, \mathbf{R}), \quad \mathbf{u} \mid \boldsymbol{\theta} \sim N_q(\mathbf{0}, \mathbf{G}), \quad (\boldsymbol{\beta}, \boldsymbol{\theta}) \sim \pi. \quad (32.6.1)$$

Level one is a linear model with known coefficients; level two is the prior on $\mathbf{u}$ that the experiment's sampling of levels supplies; level three is the part that is genuinely a matter of choice.

Proposition 32.6.1. BLUP as a posterior mean, REML as a marginal posterior

Consider [Eq. 32.6.1](#) with $\mathbf{G}$ and $\mathbf{R}$ positive definite and $\text{rank}(\mathbf{X}) = p$.

- (a) (*BLUP is a posterior mean.*) Fix $\boldsymbol{\theta}$ and take the improper flat prior $\pi(\boldsymbol{\beta}) \propto 1$. Then the joint posterior of $(\boldsymbol{\beta}, \mathbf{u})$ is

$$\begin{pmatrix} \boldsymbol{\beta} \\ \mathbf{u} \end{pmatrix} \mid \mathbf{y} \sim N_{p+q}\left(\begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{pmatrix}, \mathbf{C}^{-1}\right),$$

with $\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}}$ and $\mathbf{C}$ as in Theorem 32.3.2. In particular $E(\mathbf{u} \mid \mathbf{y}) = \hat{\mathbf{u}}$ is the BLUP, $E(\boldsymbol{\beta} \mid \mathbf{y}) = \hat{\boldsymbol{\beta}}$ is the generalized least squares estimate, and $\text{Cov}(\mathbf{u} \mid \mathbf{y})$ is the prediction error covariance of [Eq. 32.3.2](#).

- (b) (*REML is a marginal posterior.*) With the flat prior on $\boldsymbol{\beta}$ and any prior $\pi(\boldsymbol{\theta})$, the marginal posterior of the variance components is

$$\pi(\boldsymbol{\theta} \mid \mathbf{y}) \propto \pi(\boldsymbol{\theta}) \exp\{\ell_R(\boldsymbol{\theta})\}, \quad (32.6.2)$$

with ℓ_R the restricted log-likelihood of Definition 32.4.1. The REML estimate is therefore the mode of the marginal posterior under a flat prior on $\boldsymbol{\theta}$.

- (c) (*What the plug-in leaves out.*) Under (a) with $\boldsymbol{\theta}$ replaced by an estimate, the reported posterior variance is $\text{Cov}(\mathbf{u} \mid \mathbf{y}, \hat{\boldsymbol{\theta}})$, whereas the correct marginal posterior variance is $E\{\text{Cov}(\mathbf{u} \mid \mathbf{y}, \boldsymbol{\theta}) \mid \mathbf{y}\} + \text{Cov}\{E(\mathbf{u} \mid \mathbf{y}, \boldsymbol{\theta}) \mid \mathbf{y}\}$. The plug-in omits the second term, so it is too small.

Proof.

- (a) With $\pi(\boldsymbol{\beta}) \propto 1$ and $\boldsymbol{\theta}$ fixed, the joint density of $(\mathbf{y}, \mathbf{u})$ times the prior is proportional to $\exp\{-\frac{1}{2}Q(\boldsymbol{\beta}, \mathbf{u})\}$ with Q the penalized criterion [Eq. 32.3.6](#), because $-2 \log f(\mathbf{y} \mid \boldsymbol{\beta}, \mathbf{u}) =$

$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u})^\top \mathbf{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u})$ and $-2\log f(\mathbf{u}) = \mathbf{u}^\top \mathbf{G}^{-1}\mathbf{u}$, up to terms free of $(\boldsymbol{\beta}, \mathbf{u})$. By Theorem 32.3.2(b), Q is a quadratic in $\mathbf{w} = (\boldsymbol{\beta}^\top, \mathbf{u}^\top)^\top$ with positive definite matrix $\mathbf{C}$, minimized at $\hat{\mathbf{w}} = (\hat{\boldsymbol{\beta}}^\top, \hat{\mathbf{u}}^\top)^\top$, so $Q(\mathbf{w}) = (\mathbf{w} - \hat{\mathbf{w}})^\top \mathbf{C}(\mathbf{w} - \hat{\mathbf{w}}) + Q(\hat{\mathbf{w}})$. The posterior is therefore $N_{p+q}(\hat{\mathbf{w}}, \mathbf{C}^{-1})$, and Theorem 32.3.2(c) identifies the blocks.

- (b) The joint posterior is $\pi(\boldsymbol{\beta}, \boldsymbol{\theta} \mid \mathbf{y}) \propto \pi(\boldsymbol{\theta}) \exp\{\ell(\boldsymbol{\beta}, \boldsymbol{\theta})\}$ with ℓ the marginal log-likelihood Eq. 32.4.1, the random effects having already been integrated out in Eq. 32.2.2. Completing the square as in Lemma 32.4.1(b),

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{P}\mathbf{y} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}),$$

so integrating over $\boldsymbol{\beta} \in \mathbb{R}^p$ contributes the factor $(2\pi)^{p/2} \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1/2}$ and

$$\int e^{\ell(\boldsymbol{\beta}, \boldsymbol{\theta})} d\boldsymbol{\beta} = \exp\left[-\frac{1}{2}\{(n-p)\log 2\pi + \log \det \mathbf{V} + \log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) + \mathbf{y}^\top \mathbf{P}\mathbf{y}\}\right].$$

By Eq. 32.4.5 the exponent is $\ell_R(\boldsymbol{\theta})$ up to the additive constant $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{X})$, which does not involve $\boldsymbol{\theta}$. That is Eq. 32.6.2.

- (c) This is the decomposition of a variance into the mean of the conditional variance plus the variance of the conditional mean (Proposition 2.5.4), applied to the posterior distribution with $\boldsymbol{\theta}$ as the conditioning variable. The second term is nonnegative definite. ■

Part (b) is Harville's (1974) observation, and it settles a question that the frequentist derivation leaves slightly mysterious. REML looked like a technical device for recovering lost degrees of freedom; it is in fact the exact marginal likelihood of the variance components once the fixed effects have been integrated out with a flat prior. The " $n-p$ " is not a correction but a consequence.

Idea. Three readings of the same computation. Frequentist: $\hat{\mathbf{u}}$ minimizes prediction mean squared error among unbiased linear predictors. Penalized: it minimizes a sum of squares with a ridge penalty on $\mathbf{u}$. Bayesian: it is the posterior mean of $\mathbf{u}$ under a normal prior with covariance $\mathbf{G}$. Henderson's equations compute all three at once, and the three readings differ only in what they say about $\mathbf{G}$: a covariance to be estimated, a penalty to be tuned, or a prior to be chosen.

32.6.2 Priors for variance components

The conditional distributions in Eq. 32.6.1 make the conjugate choices obvious. Given $\mathbf{u}$ and $\boldsymbol{\beta}$, the sum of squares $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}\|^2$ is an ordinary normal likelihood for σ^2 , so an inverse gamma prior is conjugate; given $\mathbf{u}$, the sum $\|\mathbf{u}\|^2$ plays the same role for σ_a^2 . This is what makes the Gibbs sampler below so short. What is not obvious is which inverse gamma.

Warning. The prior $\sigma_a^2 \sim \text{IG}(\epsilon, \epsilon)$ with ϵ small is often described as non-informative. It is not, and the limit does not exist. The marginal (restricted) likelihood $\exp\{\ell_R\}$ is continuous and *strictly positive* at $\sigma_a^2 = 0$, because $\sigma_a^2 = 0$ is a perfectly good model. Any prior density whose integral diverges at the origin, such as $\pi(\sigma_a^2) \propto 1/\sigma_a^2$ — equivalently a flat prior on

$\log \sigma_a$ — therefore gives an improper posterior: $\int \pi(\sigma_a^2) e^{\ell_R} d\sigma_a^2 = \infty$. For small ϵ the $\text{IG}(\epsilon, \epsilon)$ posterior is proper but arbitrarily close to improper, and with few groups the answer depends on ϵ . Gelman (2006) made this point and recommended instead a prior that is flat or half- t on the *standard deviation* σ_a , which has finite mass near zero.

A convenient weakly informative default is the half-Cauchy on σ_a with scale A , chosen large relative to the plausible size of a group effect. It keeps the Gibbs sampler conjugate through the scale-mixture representation

$$\sigma_a^2 \mid \xi \sim \text{IG}(\tfrac{1}{2}, 1/\xi), \quad \xi \sim \text{IG}(\tfrac{1}{2}, 1/A^2),$$

whose marginal for σ_a is half-Cauchy(0, A). The auxiliary variable ξ is sampled alongside everything else.

32.6.3 A Gibbs sampler in a dozen lines

For the one-way model with a flat prior on μ , the four full conditionals are all standard. Given $\mu, \sigma^2, \sigma_a^2$, the effects are independent with

$$a_k \mid \text{rest} \sim N\left(\frac{(m/\sigma^2)(\bar{y}_k - \mu)}{m/\sigma^2 + 1/\sigma_a^2}, \frac{1}{m/\sigma^2 + 1/\sigma_a^2}\right),$$

which is Eq. 32.3.3 with the current parameter values; given the effects, $\mu \sim N(\overline{y - \mathbf{Za}}, \sigma^2/n)$; and the two variances have inverse gamma conditionals with shapes $n/2$ and $g/2$ plus their prior shapes. Nothing needs tuning.

```
import numpy as np

g, m = 12, 4
n = g * m
mu, sigma_a, sigma = 2.50, 0.20, 0.25          # the values used to simulate

rng = np.random.default_rng(20320048)
lab_effect = sigma_a * rng.standard_normal(g)
y = mu + np.repeat(lab_effect, m) + sigma * rng.standard_normal(n)

Y = y.reshape(g, m)                          # the REML fit of Section 32.4
ms_between = m * np.sum((Y.mean(axis=1) - Y.mean()) ** 2) / (g - 1)
ms_within = np.sum((Y - Y.mean(axis=1)[:, None]) ** 2) / (n - g)
reml_s2, reml_s2a = ms_within, (ms_between - ms_within) / m

def gibbs(y, g, m, draws=40000, burn=2000, prior="inverse-gamma", A=1.0, seed=0):
    """Posterior draws of (mu, sigma^2, sigma_a^2, a) with a flat prior on mu."""
    rng = np.random.default_rng(seed)
    Y = y.reshape(g, m)
    n = g * m
    mu_, s2, s2a, xi = Y.mean(), Y.var(), 0.1, 1.0
    out = np.empty((draws, 3 + g))
    for t in range(draws + burn):
        prec = m / s2 + 1 / s2a                # a_k | rest
        a = rng.normal((m / s2) * (Y.mean(axis=1) - mu_) / prec, np.sqrt(1 / prec))
        mu_ = rng.normal((y - np.repeat(a, m)).mean(), np.sqrt(s2 / n))    # mu |
        rest
        resid = y - mu_ - np.repeat(a, m)
        s2 = 1 / rng.gamma(0.001 + n / 2, 1 / (0.001 + resid @ resid / 2))
        if prior == "inverse-gamma":          # sigma_a^2 | rest
```

```

s2a = 1 / rng.gamma(0.001 + g / 2, 1 / (0.001 + a @ a / 2))
else:
    s2a = 1 / rng.gamma((g + 1) / 2, 1 / (1 / xi + a @ a / 2))
    xi = 1 / rng.gamma(1.0, 1 / (1 / A**2 + 1 / s2a))
if t >= burn:
    out[t - burn] = np.concatenate([[mu_, s2, s2a], a])
return out

```

[Run this code online](#)

```

# a shorter run than the 40 000 draws quoted in the text, so the cell finishes
quickly
short = gibbs(y, g, m, draws=4000, burn=500, prior="half-Cauchy", seed=12)
print(f"posterior median of sigma_a^2 {np.median(short[:, 2]):.4f}")
print(f"posterior mean of mu {short[:, 0].mean():.4f}")
print(f"REML estimate of sigma_a^2 {reml_s2a:.4f}")

```

[Run this code online](#)

Example. The proficiency study, four ways

Run the sampler for 40 000 draws on the data of Example (A laboratory proficiency study). Under the $IG(0.001, 0.001)$ prior the posterior median of σ_a^2 is 0.0356; under the half-Cauchy prior with $A = 1$ it is 0.0438. The REML estimate 0.0386 sits between them. The two priors therefore disagree by about a fifth of the estimate, with twelve laboratories — which is already a reason to report the prior.

The posterior means of the twelve effects track the BLUPs closely: their correlation exceeds 0.99 and the largest in absolute value is 0.294 against the BLUP's 0.321, slightly more shrunk — not because the posterior sits below the REML estimate, since its mean 0.0431 is above it, but because the shrinkage factor $B = m\sigma_a^2/(\sigma^2 + m\sigma_a^2)$ is concave in σ_a^2 : averaging B over a posterior with mass near zero weights the data less than $\hat{\sigma}_a^2$ does. Their posterior standard deviations average 0.1218, against the plug-in prediction standard error 0.1156 of Eq. 32.3.4: 5 per cent wider, which is the uncertainty about σ_a^2 that the plug-in throws away, exactly as Proposition 32.6.1(c) says. The same happens for the fixed effect: the posterior standard deviation of μ is 0.0709, against the plug-in $\sqrt{\hat{\lambda}_1/n} = 0.0672$.

Panel (b) of Figure 32.6.1 repeats the analysis on the first 5 laboratories. The posterior medians are now 0.0435 and 0.0693, a difference of sixty per cent, against a REML estimate of 0.0493. With five groups the prior is doing a large part of the work, and saying so is not optional.

32.6.4 What the Bayesian version buys, and what it costs

Three gains. The uncertainty about θ is carried through automatically, so intervals for fixed effects and prediction intervals for random effects are honestly wider, with no need for the corrections of Section 32.5. The boundary problem disappears: the posterior of σ_a^2 is a proper distribution on $[0, \infty)$, with no point mass and no chi-bar-squared mixture to calibrate. And the same machinery handles models with no closed form, which is what Chapter 33 needs for richer covariance structures and Chapter 40 needs once the response is not normal.

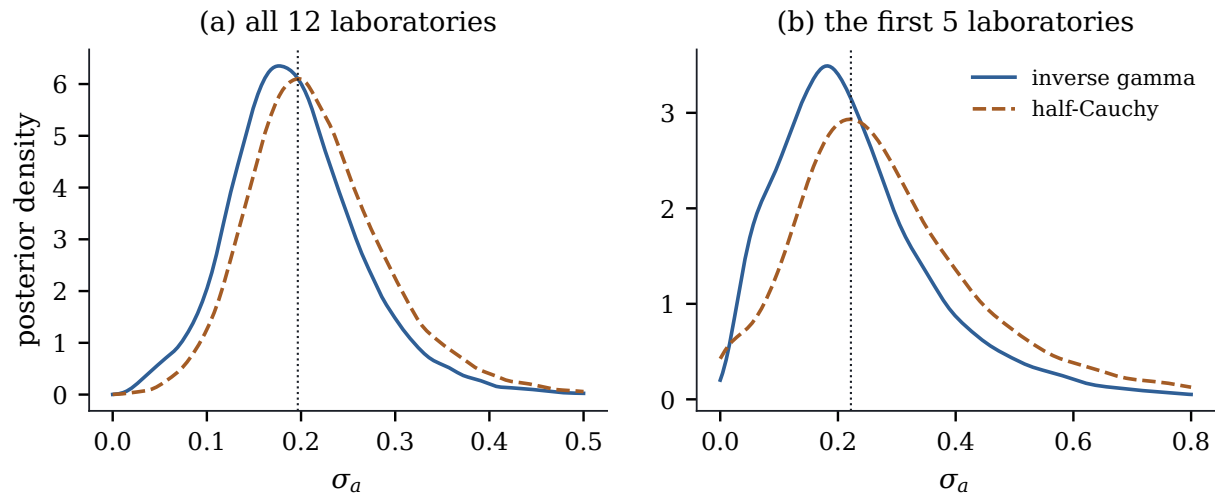


Figure 32.6.1: Posterior densities of the between-laboratory standard deviation σ_a under two priors, from a Gibbs sampler with 40 000 draws. The dotted line marks the REML estimate. (a) All twelve laboratories. (b) The first five, where the prior matters a great deal.

Two costs. A prior for the variance components must be chosen, and with few groups it matters, as panel (b) shows; the only defensible practice is to state it and show the sensitivity. And the answers are not exactly the frequentist ones: the two frameworks agree on the point estimates, but they part company on the spread, and the frequentist coverage of a credible interval for a variance component is not guaranteed (Proposition 12.6.4).

32.6.5 Exercises

32.6.5.1 A. Check your understanding

Exercise. A1

Explain in two sentences why a random effect and a prior are formally the same object, and name one thing that distinguishes them in practice.

32.6.5.2 B. Practice

Exercise. B1

Show that in the one-way model the marginal posterior density of σ_a^2 under the prior $\pi(\sigma_a^2) \propto \sigma_a^{-2}$ is not integrable, by showing that $\exp\{\ell_R(\sigma_a^2)\}$ tends to a strictly positive limit as $\sigma_a^2 \downarrow 0$ with σ^2 profiled out. Contrast with the behaviour of the prior $\pi(\sigma_a) \propto 1$.

Solution. By Eq. 32.4.8, $-2\ell_R$ is continuous in (λ_1, λ_2) on $\lambda_1 \geq \lambda_2 > 0$, and at $\sigma_a^2 = 0$ it equals $(n-1)\log \lambda_2 + (\text{SSB} + \text{SSE})/\lambda_2$, finite and minimized at a positive λ_2 . So $\exp\{\ell_R\} \rightarrow c > 0$ as $\sigma_a^2 \downarrow 0$, and $\int_0^\delta c \sigma_a^{-2} d\sigma_a^2 = \infty$. With $\pi(\sigma_a) \propto 1$ the Jacobian gives $\pi(\sigma_a^2) \propto \sigma_a^{-1}$, and $\int_0^\delta \sigma_a^{-1} d(\sigma_a^2) = 2 \int_0^{\sqrt{\delta}} d\sigma_a < \infty$: the posterior is proper near the origin.

Exercise. B2

In Proposition 32.6.1(c), suppose θ takes only two values with posterior probabilities $1/2$ each, giving conditional posterior means $\hat{u}_1, \hat{u}_2$ and common conditional variance v for a coordinate. Compute the correct marginal posterior variance, and say how large the disagreement between $\hat{u}_1$ and $\hat{u}_2$ must be before the plug-in understates the standard deviation by ten per cent.

32.6.5.3 C. Going deeper

Exercise. C1

Formula Eq. 32.6.2 shows REML is the posterior mode of θ under a flat prior on θ . A mode is not invariant to reparameterization: the mode of the posterior of σ_a^2 is not the square of the mode of the posterior of σ_a . Reconcile this with Theorem 32.4.2(c), which says the REML estimate is well defined. What is invariant, and what is not?

Exercise. C2

Take the balanced one-way model with σ^2 known, $m = 1$ so that $\bar{y}_k = y_k$, and μ known to be zero. Show that plugging the unbiased moment estimate of σ_a^2 into Eq. 32.3.3 gives a shrinkage factor of the form $1 - c\sigma^2 / \sum_k y_k^2$, and compare with the James–Stein estimator of Theorem 27.5.2. Which constant c does the plug-in give, and which does James–Stein use?

Solution. With $m = 1$ and $\mu = 0$ the data are $y_k \sim N(a_k, \sigma^2)$ with $a_k \sim N(0, \sigma_a^2)$, so marginally $y_k \sim N(0, \sigma^2 + \sigma_a^2)$. The moment estimate from $\sum_k y_k^2$ is $\hat{\sigma}_a^2 = \sum_k y_k^2 / g - \sigma^2$, and

$$B = \frac{\hat{\sigma}_a^2}{\hat{\sigma}_a^2 + \sigma^2} = 1 - \frac{g\sigma^2}{\sum_k y_k^2},$$

so $c = g$ in the stated form. The James–Stein estimator of Theorem 27.5.2 uses $1 - (g - 2)\sigma^2 / \sum_k y_k^2$. The difference is the same degrees-of-freedom correction that separates ML from REML: $g - 2$ is the constant that makes $(g - 2)\sigma^2 / \sum y_k^2$ unbiased for $\sigma^2 / (\sigma^2 + \sigma_a^2)$, and it is what makes the estimator dominate the unshrunk one for every α , not merely on average.